INTERNET
SOFTWARE ARCHITECTURE

Weather App Report
Individual Assessment

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Contents

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Introduction

This is a real-time web-based weather application following a Client-server architecture made by using HTML, CSS, JS, PHP, & MySQL. This version represents Prototype 3 and hosted using InfinityFree, making it accessible over internet without the need for a local server environment.

The objective of this prototype was to store data on local storage and deploy using a free hosting platform. Currently this system retrieves from OpenWeather API, stores data and present it in a user-friendly dashboard.

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Strengths

As the application follows Client-server architecture and is service oriented, and hosted on InfinityFree using free account which enables real world accessibility and shows the process of deploying a web application. The application uses OpenWeather API to fetch data and to reduce excessive API call, improved performance, retrieved data is cached on both server side and within the browser. This helps to manage API limits and faster response time.

A key strength of this prototype is the implementation of offline access using local storage. When the user is offline, the application retrieves and display cached data without connecting to server or external API. This helps to enhance user experience during network outages.

This application can display weather details such as city name, date, temperature, humidity, pressure, wind speed, wind direction, and weather conditions with a clear and informative error messages are displayed when invalid city names are entered, cached data is unavailable, or connectivity issues occur.

Weakness

Even after successful deployment, the application is highly dependent on the InfinityFree hosting platform. If InfinityFree gets downtime or service disruption, application becomes unavailable due the PHP backend and database are hosted on the same server creating a single point of failure, directly affecting system availability. This application is hosted on free InfinityFree account which comes with limitations. According to InfinityFree's suspension policy, exceeding 50,000 daily hits or high resource usage results in a 24-hour suspension, during which the website becomes completely inaccessible.

Even offline access is supported, it has some functional limitations. The application can only display weather data for cities that were previously searched and stored in localStorage. New city searches cannot be performed while offline and data is stored in browser mean cached data is lost if browser data is cleared or a different device is used.

The application also depends on the **OpenWeather API** free plan, which comes with request limits. Once the API free call is reached, new live weather data cannot be fetched until the limit resets and this application doesn't support authentication or authorization, making it vulnerable to misuse, automated traffic, or malicious requests that could unintentionally trigger hosting limits or suspension.

Conclusion & Future scope

This prototype successfully shows the transition from a locally developed weather application to a deployed, publicly accessible system. This application has some limitations associated with free hosting, API call and authentication.

In future this application could include migrating the application to a paid or cloud-based hosting service, implementing user authentication and security mechanisms, introducing rate limiting, supporting favorite city management, enabling live location-based weather detection, and implementing advanced offline capabilities using service workers

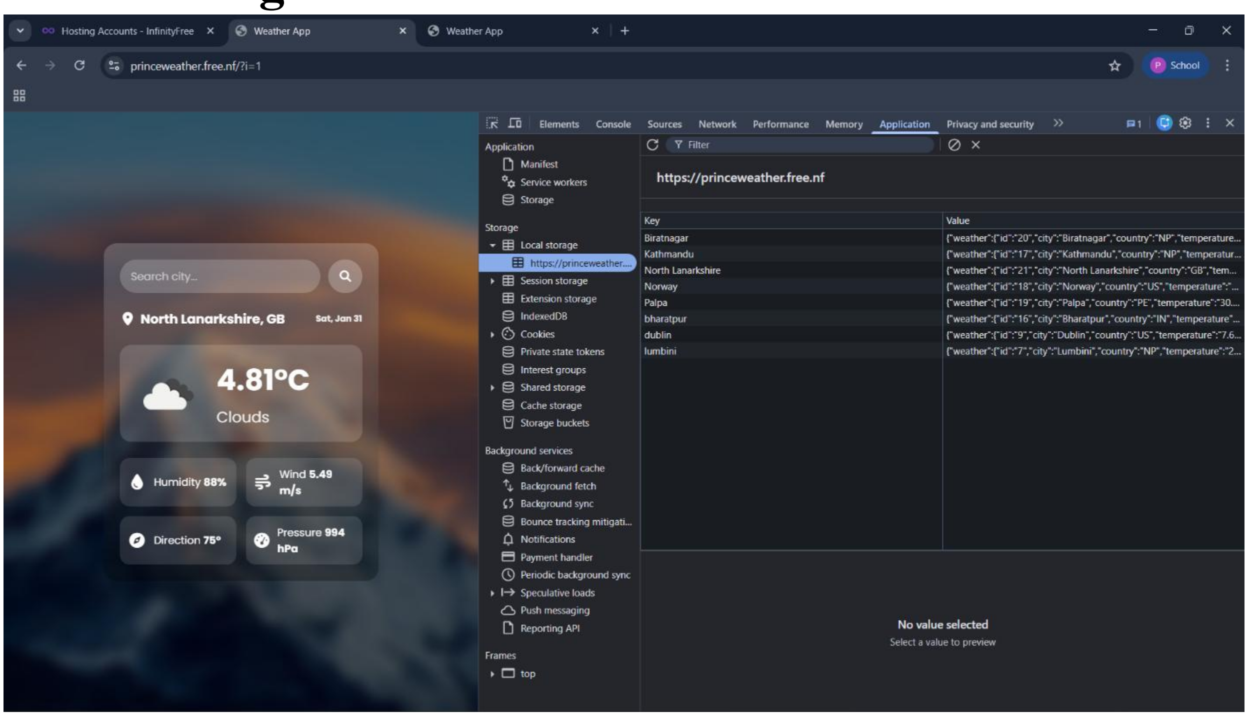
Link for application:

https://princeweather.free.nf/?i=1

Link for Youtube video:

https://youtu.be/lOjUcRijlgI

Local storage:



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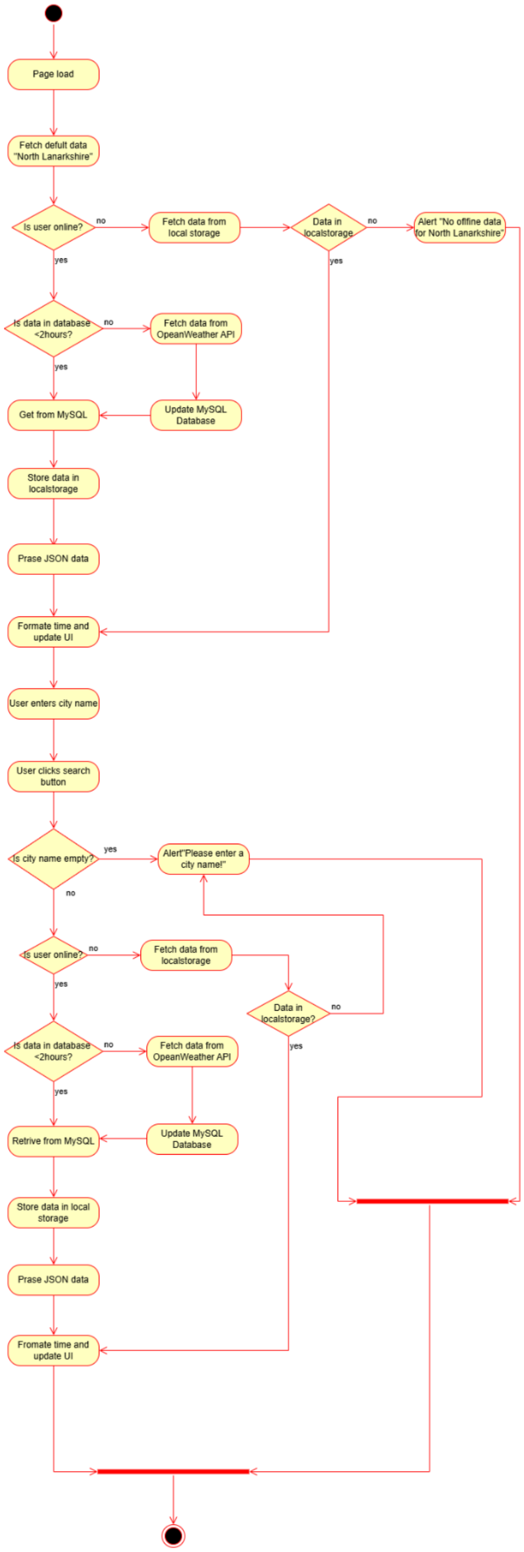
UML Diagram

1. Activity Diagram



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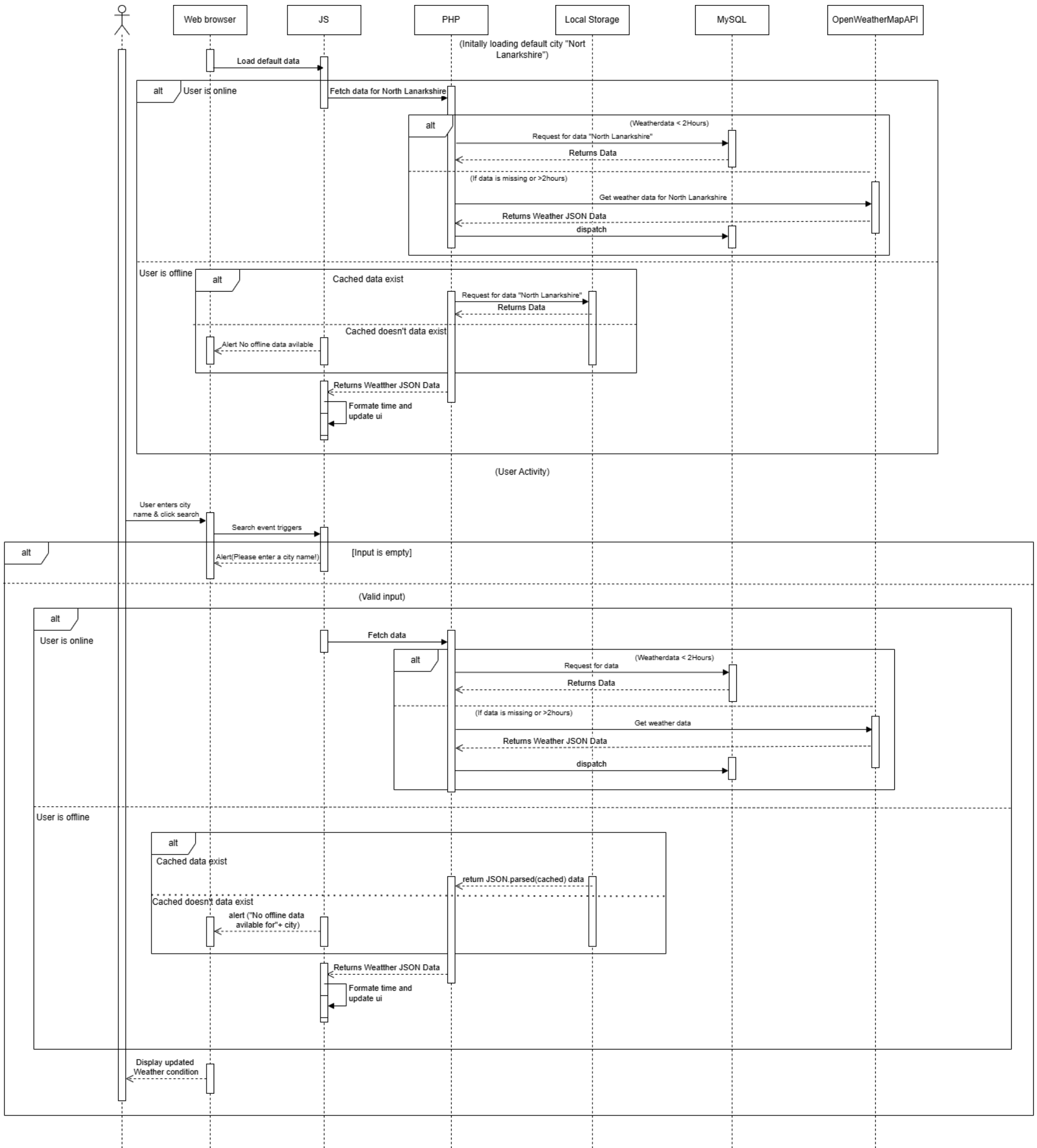
2. Sequence Diagram



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3. Deployment Diagram